

SSILP Math

Beginner

LOGARITHMIC FUNCTIONS



Learning Objectives

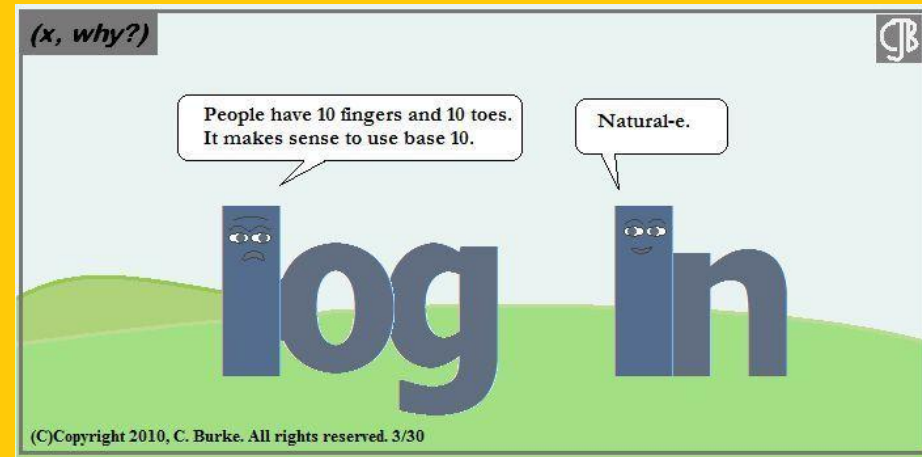
- Understand the concept of logarithmic functions and their relationship with exponential functions
- Learn how to rewrite logarithmic equations as exponential equations and vice versa
- Evaluate logarithmic expressions and understand the properties of logarithms.
- Explore the graphs of logarithmic functions and understand their domain and range.
- Work with natural logarithms and common logarithms.
- Solve word problems involving logarithmic functions.

Session Outline

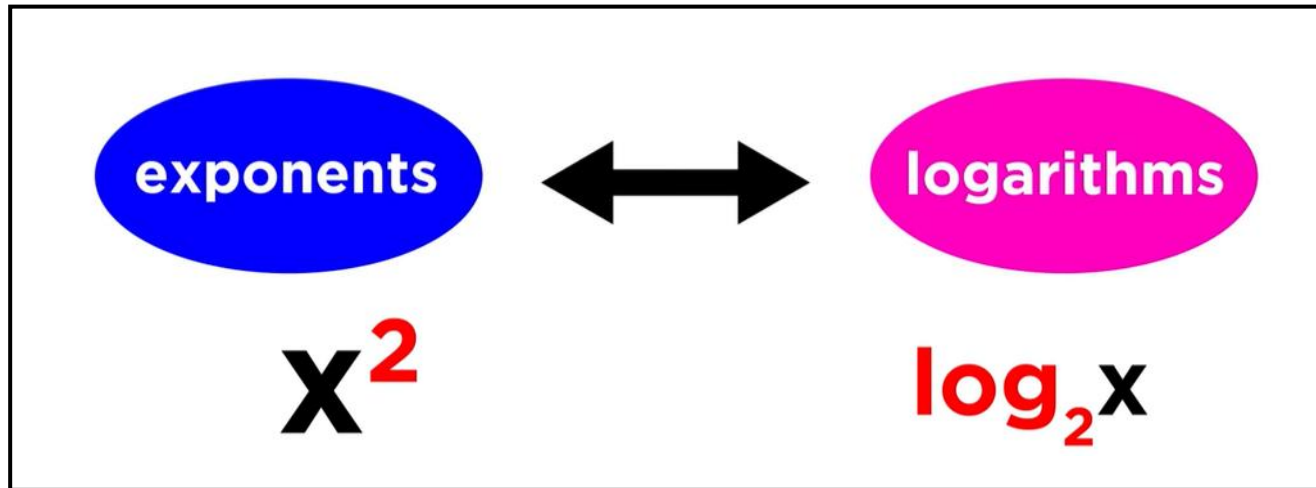
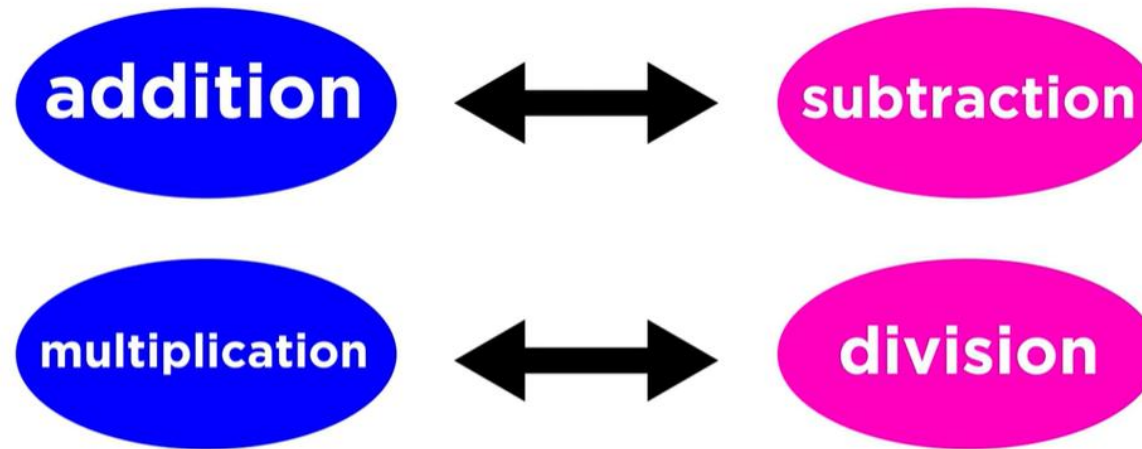
1. Logarithmic functions



1. Logarithmic functions



Inverse operations



Logarithmic form vs Exponential form

$$\log_a x = y$$

logarithms can express
exponential relationships


$$\log_b x = y$$

“log base b of x equals y”

$$b^y = x$$

Example

$$\log_2 8 = 3$$

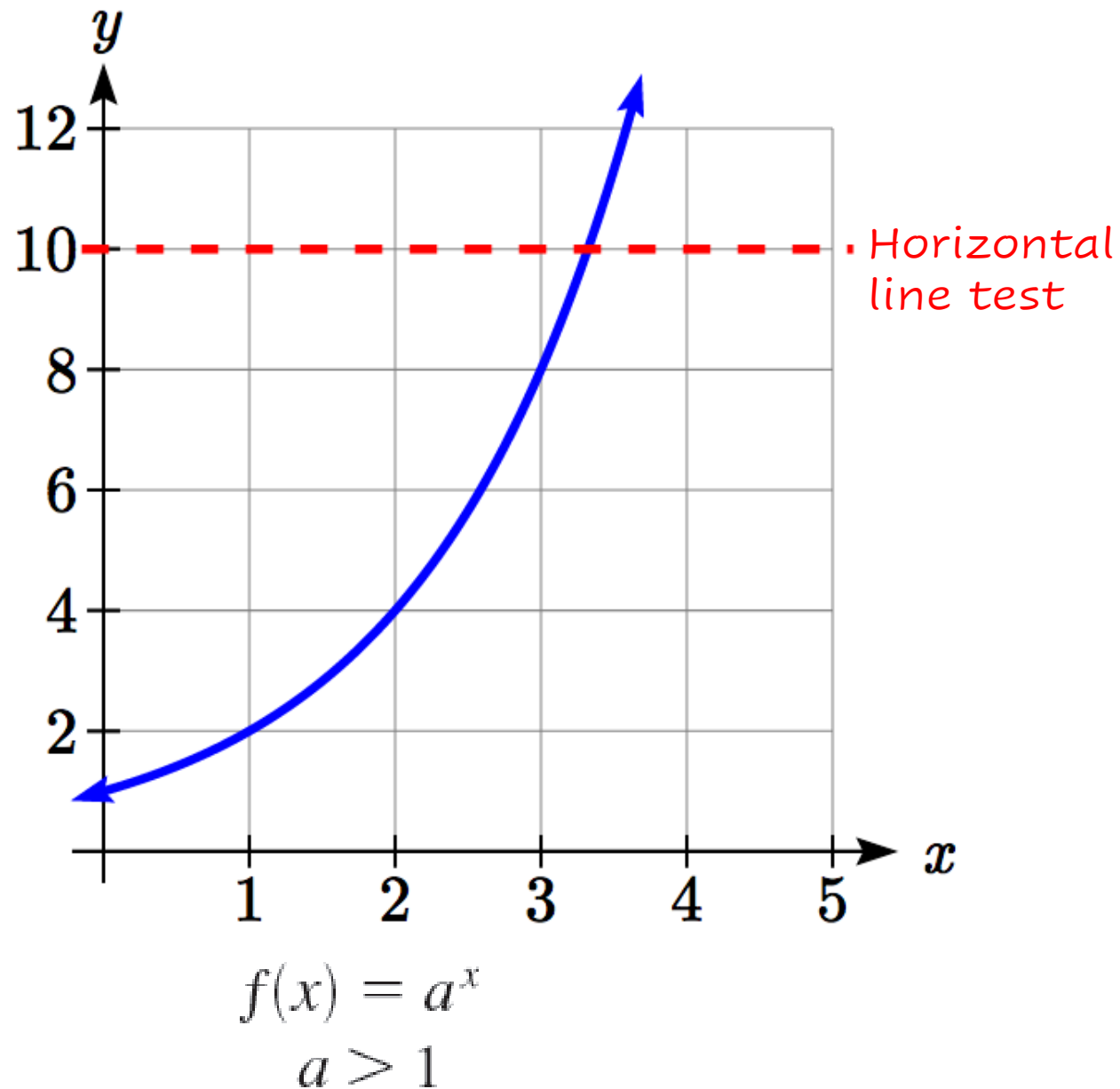
$$2^3 = 8$$

Example

$$\log_5 25 = 2$$

$$5^2 = 25$$

Logarithm function: Inverse of exponential function



Exponential function is a
one-to-one
function

→ Has an inverse function

The inverse of an
exponential function with base b
is
logarithmic function with base b .

Logarithms are exponents!

$$y = \log_b x$$

and

$$x = b^y$$

Equivalent statements

y is the exponent (from statement 2)

→ logarithm is also an exponent (from statement 1)

Definition of Logarithmic function

DEFINITION OF THE LOGARITHMIC FUNCTION

Let a be a positive number with $a \neq 1$. The **logarithmic function with base a** , denoted by **$\log_a$** , is defined by

$$\log_a x = y \quad \Leftrightarrow \quad a^y = x$$

So $\log_a x$ is the *exponent* to which the base a must be raised to give x .

- a) Rewrite $\log_3 243 = 5$ as an exponential equation.
- b) Rewrite $2^6 = 64$ as a logarithmic equation.

1.1 Rewrite a Logarithms and Exponentials

Rewrite a Logarithms and Exponentials

- a) Rewrite $\log_3 243 = 5$ as an exponential equation.
- b) Rewrite $2^6 = 64$ as a logarithmic equation.

Solution

- a) Since logarithms equal exponents, 5 is the exponent.

The base of the logarithm is also the base of the exponential, so 3 is the base.

So, $3^5 = 243$.

- a) Since logarithms equal exponents, the logarithm will equal 6.

The base of the logarithm is the same as the base of the exponential, so the base is 2.

So, $\log_2 64 = 6$

Evaluating Logarithms

(a) $\log_{10} 1000 = 3$ because $10^3 = 1000$

(b) $\log_2 32 = 5$ because $2^5 = 32$

(c) $\log_{10} 0.1 = -1$ because $10^{-1} = 0.1$

(d) $\log_{16} 4 = \frac{1}{2}$ because $16^{1/2} = 4$

EXAMPLE

Evaluate

a) $\log_2 8$

b) $\log_3 81$

c) $\log_{1/2} 16$

d) $\log_4 \frac{1}{16}$

1.2 Evaluating Logarithms

a) $\log_2 8$

Logarithms are exponents, **so we need to rewrite the expression as an exponential**. Since $2^3 = 8$, the answer is 3.

b) $\log_3 81$

Since $3^4 = 81$, the answer is 4.

c) $\log_{1/2} 16$

- Since $\left(\frac{1}{2}\right)^{-4} = 16$, the answer is -4 .

d) $\log_4 \frac{1}{16}$

- Since $4^{-2} = \frac{1}{16}$, the answer is -2

Properties of Logarithms

PROPERTIES OF LOGARITHMS

Property

$$1. \log_a 1 = 0$$

$$2. \log_a a = 1$$

$$3. \log_a a^x = x$$

$$4. a^{\log_a x} = x$$

Reason

We must raise a to the power 0 to get 1.

We must raise a to the power 1 to get a .

We must raise a to the power x to get a^x .

$\log_a x$ is the power to which a must be raised to get x .

These are based on the inverse function property!

Inverse Function Property:

$$f^{-1}(f(x)) = x$$

$$f(f^{-1}(x)) = x$$

Examples

$$\log_5 1 = 0$$

Property 1

$$\log_5 5^8 = 8$$

Property 3

$$\log_5 5 = 1$$

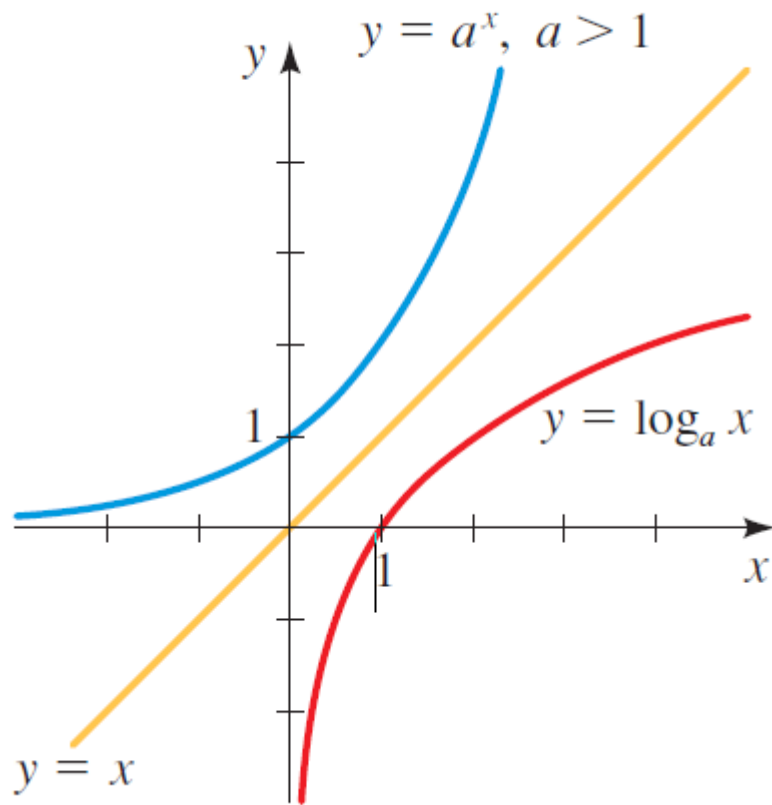
Property 2

$$5^{\log_5 12} = 12$$

Property 4

Graphs of Logarithmic Functions

Domain and range of Log functions



Recall that if a one-to-one function f has domain A and range B , then its inverse function f^{-1} has domain B and range A . Since the exponential function $f(x) = a^x$ with $a \neq 1$ has domain $\mathbb{R}$ and range $(0, \infty)$, we conclude that its inverse function, $f^{-1}(x) = \log_a x$, has domain $(0, \infty)$ and range $\mathbb{R}$.

The graph of $f^{-1}(x) = \log_a x$ is obtained by reflecting the graph of $f(x) = a^x$ in the line $y = x$. Figure 2 shows the case $a > 1$. The fact that $y = a^x$ (for $a > 1$) is a very rapidly increasing function for $x > 0$ implies that $y = \log_a x$ is a very slowly increasing function for $x > 1$ (see Exercise 102).

Since $\log_a 1 = 0$, the x -intercept of the function $y = \log_a x$ is 1. The y -axis is a vertical asymptote of $y = \log_a x$ because $\log_a x \rightarrow -\infty$ as $x \rightarrow 0^+$.

Domain and range of a Logarithmic function

Sketch the graph of each function. State the domain, range, and asymptote.

(a) $g(x) = -\log_2 x$ (b) $h(x) = \log_2(-x)$

SOLUTION

- (a) We start with the graph of $f(x) = \log_2 x$ and reflect in the x -axis to get the graph of $g(x) = -\log_2 x$ in Figure 5(a). From the graph we see that the domain of g is $(0, \infty)$, the range is the set $\mathbb{R}$ of all real numbers, and the line $x = 0$ is a vertical asymptote.
- (b) We start with the graph of $f(x) = \log_2 x$ and reflect in the y -axis to get the graph of $h(x) = \log_2(-x)$ in Figure 5(b). From the graph we see that the domain of h is $(-\infty, 0)$, the range is the set $\mathbb{R}$ of all real numbers, and the line $x = 0$ is a vertical asymptote.

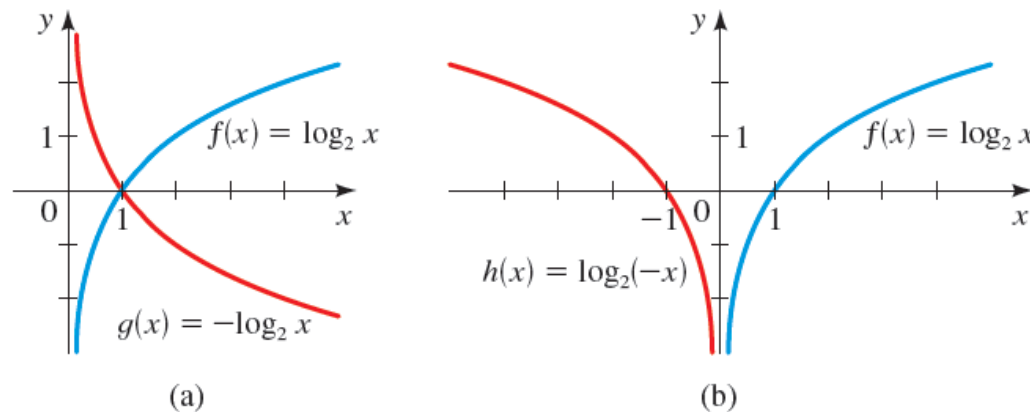
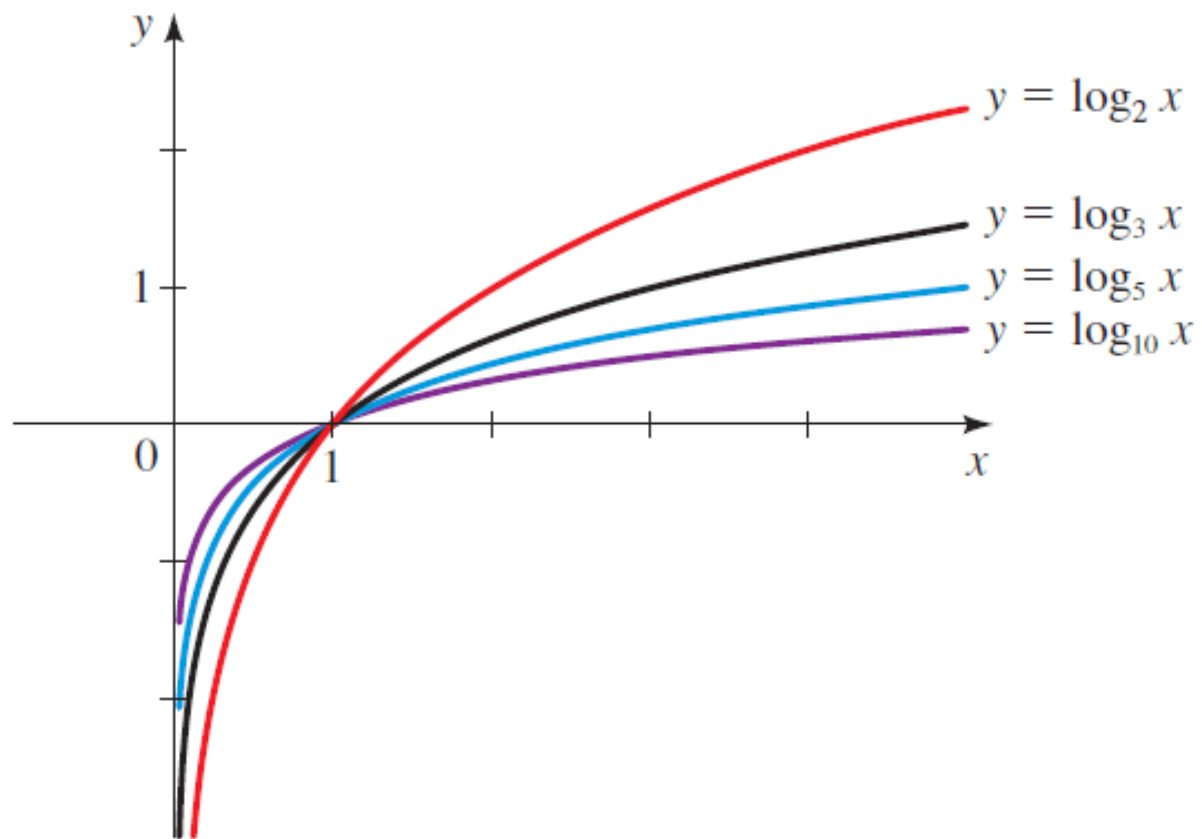


FIGURE 5

A family of logarithmic functions



These graphs are drawn by reflecting the graphs of

$$y = 2^x$$

$$y = 3^x$$

$$y = 5^x$$

$$y = 10^x$$

Across the line $y = x$

Special Logarithms

Two **special** logarithms
commonly used in real-world
applications.

logarithm **base 10** called the
common log
written without the base like
 $y = \log x$

logarithm **base e** called the
natural log
written as
 $y = \ln x$

Common Logarithm

COMMON LOGARITHM

The logarithm with base 10 is called the **common logarithm** and is denoted by omitting the base:

$$\log x = \log_{10} x$$

Values of the common log

number	number as exponential	log(number)
1000	10^3	3
100	10^2	2
10	10^1	1
1	10^0	0
0.1	10^{-1}	-1
0.01	10^{-2}	-2
0.001	10^{-3}	-3

Note the Notation!

This notation ($\log x$) is not universal!

Some mathematicians use $\log x = \ln x$

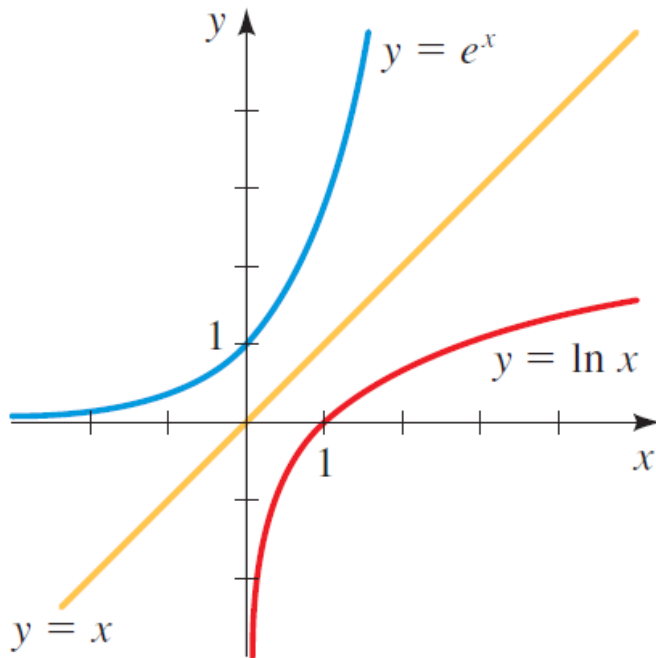
Natural Logarithm

NATURAL LOGARITHM

The logarithm with base e is called the **natural logarithm** and is denoted by **ln**:

$$\ln x = \log_e x$$

$$\ln x = y \iff e^y = x$$



The number e is the most convenient choice for the base for logarithms, for the purpose of calculus

Properties of Natural logarithms

PROPERTIES OF NATURAL LOGARITHMS

Property

Reason

1. $\ln 1 = 0$

We must raise e to the power 0 to get 1.

2. $\ln e = 1$

We must raise e to the power 1 to get e .

3. $\ln e^x = x$

We must raise e to the power x to get e^x .

4. $e^{\ln x} = x$

$\ln x$ is the power to which e must be raised to get x .

These are similar to the
properties of logarithms we saw
earlier

Evaluating the Natural Logarithm Function

(a) $\ln e^8 = 8$

Definition of natural logarithm

(b) $\ln\left(\frac{1}{e^2}\right) = \ln e^{-2} = -2$

Definition of natural logarithm

EXAMPLE

Evaluate the following expressions

(a) $e^{\ln \sqrt{3}}$

(b) $e^{\ln(1/\pi)}$

(c) $\ln(1/e)$

ACTIVITY 1.3

1.3 Evaluate the following expressions

(a) $e^{\ln \sqrt{3}}$

$$\begin{aligned} * e^{\ln(x)} &= x \\ e^{\ln \sqrt{3}} &= \sqrt{3} \end{aligned}$$

(b) $e^{\ln(1/\pi)}$

$$\begin{aligned} * e^{\ln(x)} &= x \\ e^{\ln(\frac{1}{\pi})} &= \frac{1}{\pi} \end{aligned}$$

(c) $\ln(1/e)$

$$\ln \frac{1}{e} = \ln e^{-1} = -1$$

Finding the domain of a logarithmic function

25

(a) $f(x) = \log_5(8 - 2x)$

ACTIVITY 1.4

1.4 Finding the domain of a logarithmic function

(a) $f(x) = \log_5(8 - 2x)$

The logarithmic function $f(x) = \log_a x$ is defined only when x is greater than 0.

This means that the given function is only defined when $8 - 2x$ is greater than zero.

Thus,

$$\begin{aligned} 8 - 2x &> 0 \\ -2x &> 0 - 8 \\ -2x &> -8 \end{aligned}$$

Divide -2 to both sides of the inequality.
Note that the inequality symbol will flip to the opposite direction since a negative number was divided to each side.

$$\begin{aligned} x &< \frac{-8}{-2} \\ x &< 4 \end{aligned}$$

Therefore, the domain of the given function is $x < 4$. In interval notation, the domain is $(-\infty, 4)$.

1.4 Finding the domain of a logarithmic function

SOLUTION

- (a) **Carbon Dating** The age of an ancient artifact can be determined by the amount of radioactive carbon-14 remaining in it. If D_0 is the original amount of carbon-14 and D is the amount remaining, then the artifact's age A (in years) is given by

$$A = -8267 \ln\left(\frac{D}{D_0}\right)$$

Find the age of an object if the amount D of carbon-14 that remains in the object is 73% of the original amount D_0 .

- (b) **Bacteria Colony** A certain strain of bacteria divides every 3 hours. If a colony is started with 50 bacteria, then the time t (in hours) required for the colony to grow to N bacteria is given by

$$t = 3 \frac{\log(N/50)}{\log 2}$$

Find the time required for the colony to grow to a million bacteria.

1.5 Word problems

(a) Using $D = 0.73D_0$ we have $A = -8267 \ln\left(\frac{D}{D_0}\right) = -8267 \ln 0.73 \approx 2602$ years.

(b) Substituting $N = 1,000,000$ we get $t = 3 \frac{\log(N/50)}{\log 2} = 3 \frac{\log 20,000}{\log 2} \approx 42.86$ hours.